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'''
Write a Python program to check whether a number is even or odd.
Write a program to find the largest among three numbers.
Write a Python program to calculate factorial of a number.
Write a program to print the Fibonacci series up to N terms.
Write a Python program to reverse a string without using built-in
reverse functions.
Write a program to count vowels and consonants in a string.
Write a Python program to check whether a string is palindrome or
not.'''

# Program to find even and odd

num=int(input("Enter a number"))
if num % 2 == 0:                                # we can also use num % 2 !
    print("Even")
else:
    print("odd")

Enter a number 89098
Even

num=int(input("Enter a number"))
if num & 1:                                     # I used the bitwise
operator to check for odd or even using the last bit of the number.
    print("Odd")
else:
    print("Even")

Enter a number 56
Even

find= lambda x: "Even" if x%2==0 else "Odd"      # used the lambda
function to check for odd or even in a single line of code.
print(find(34))

Even

# Write a program to find the largest among three numbers.
a=float(input("Enter the first Number:"))
b=float(input("Enter the Second Number:"))
c=float(input("Enter the third Number:"))

""" we have to compare all the three numbers with each other.
    we will compare each one number with other"""

if a>=b and a>=c:                             #basic comparison of numbers using if-
else statements.
    print("Greatest:",a)

```

```

elif b>=a and b>=c:
    print("Greatest:",b)
else:
    print("Greatest:",c)

```

```

Enter the first Number: 56
Enter the Second Number: 89
Enter the third Number: 522

```

```

Greatest: 522.0

```

```

a=float(input("Enter the first Number:"))
b=float(input("Enter the Second Number:"))
c=float(input("Enter the third Number:"))

```

```

""" we have to compare all the three numbers with each other.
    In these 2nd approach we will use the max function"""

```

```

greatest=max(a,b,c) # max function will return the
greatest number among the three numbers.

```

```

print("Largest number is:",greatest)

```

```

Enter the first Number: 56
Enter the Second Number: 54
Enter the third Number: 85

```

```

Largest number is: 85.0

```

```

# Write a Python program to calculate factorial of a number.

```

```

def fact(n):
    if n==1 or n==0:
        return 1
    return n * fact(n-1) #It helps to calculate the
factorial of a number using recursion.

```

```

num = int(input("Enter a number:"))

```

```

print("Factorial is:",fact(num))

```

```

Enter a number: 6

```

```

Factorial is: 720

```

```

import math

```

```

num=int(input("Enter a number:"))

```

```

print("factorial of {num} is:",math.factorial(num)) # second
approach using the math module to calculate.

```

Enter a number: 8

factorial of {num} is: 40320

Write a program to print the Fibonacci series up to N terms.

```
n=int(input("Enter the number"))
```

```
a=0
```

```
b=1
```

```
print("Fibonacci Series")
```

```
for i in range(n):
```

```
    print(a,end="") # we will print the initial value of a and the last value of b.
```

```
    c=a+b
```

```
    a=b
```

```
    b=c
```

Enter the number 2

Fibonacci Series

01

```
n=int(input("Enter the Number:"))
```

```
a,b=0,1
```

```
print("Finonacci Series")
```

```
for i in range(n):
```

```
    print(a,end="")
```

```
    a,b=b,a+b # a,b is assinged the values of b and a+b each.
```

Enter the Number: 5

Finonacci Series

01123

Reverse a string without using built-in reverse functions.

```
text=input("Enter the text=")
```

```
l=list(text)
```

```
for i in range(len(l)-1,-1,-1): # we interate the list in reverse order and print each character using the -1 logic.
```

```
    print(l[i], end="")
```

Enter the text= errr

rrre

Write a program to count vowels and consonants in a string.

```
text="Pramod"
```

```
v= "aeiouAEIOU" # we assigned all the vowels to the v variable.
```

```
vowels, consonants =0,0      # at start we will have the count
                              # initialized to 0.
for char in text:            # we now iterate through the text for every
                              # character.
    if char in v:
        vowels+=1            # if the character is vowel then we will
                              # increase the count of vowels by 1.
    else:
        consonants+=1        # if the character is not vowel then we
                              # will increase the count of consonants by 1.
```

```
print(f"Vowels: {vowels}, Consonants: {consonants}")
```

```
Vowels: 2, Consonants: 4
```

```
text="Pramod"
v= "aeiouAEIOU"
vowels, consonants =0,0
vowels = sum(1 for char in text if char in v)
consonants = sum(1 for char in text if char not in v)
```

```
print(f"Vowels: {vowels}, Consonants: {consonants}")
```

```
Vowels: 2, Consonants: 4
```

```
# Write a Python program to check whether a string is palindrome or
not.'
```

```
text=input("Enter the text=")
```

```
if text== text[::-1]:        # we used the slicing method to
                              # reverse the string and check if its a
                              # palindrome or not.
    print("Palindrome")
else:
    print("Not Palindrome")
```

```
Palindrome
```

```
"""Initailly we will reverse the string using the slicing method and
then compare it with the original string to check if its a palindrome
or not.
```

```
For that we created the rev function then reverse the string and then
the the function is called to check if the string is palindrome or
not."""
```

```
def rev(text):
    reverse=""

    for i in range(len(text)-1,-1,-1):
        reverse=reverse + text[i]

    return reverse
```

```
text=input("Enter the text=")
reverse=rev(text)
print("Original text:",text)
print("Reverse:",reverse)
```

```
if text==reverse:
    print("Palindrome")
else:
    print("Not palindrome")
```

```
Original text: madam
Reverse: madam
Palindrome
```